FOUNDRY TECHNOLOGY NOTES

Cores, Core Making and Melting Equipment

with detailed study of the Cupola Furnace

Reference framework: S. K. Hajra Choudhury, A. K. Hajra Choudhury and S. K. Nirjhar Roy, Elements of Workshop Technology, Vol. I: Manufacturing Processes.

These notes are paraphrased and organized for study and examination preparation.

Topics Covered

- 1. Core - Definition and Functions
- 2. Types of Cores
- 3. Core Sand - Composition, Properties and Preparation
- 4. Core Prints
- 5. Core Boxes and Types
- 6. Melting Equipment - Classification
- 7. Cupola Furnace - Construction
- 8. Cupola Furnace - Working
- 9. Zones of a Cupola Furnace
- 10. Cupola Efficiency and Performance
- 11. Quick Revision Tables and Viva Questions

Study approach: First understand why cores are needed in a mould, then learn how cores are made and supported. After that, study how metals are melted, with special emphasis on the cupola furnace used for cast iron.

1. Core - Definition and Functions

Definition. A core is a preformed body, usually made of specially prepared sand, that is placed inside a mould to produce a cavity, hole, recess, undercut, or internal surface in the final casting. After solidification, the core is broken and removed from the casting.

In ordinary sand moulding, the pattern forms mainly the external surface of the casting. Whenever the required shape cannot be produced by the mould cavity alone, a core is introduced.

Main functions of a core

- To produce holes and internal cavities in castings.
- To produce recesses, pockets and complicated internal passages.
- To form portions of a casting that cannot conveniently be made by the pattern because of undercuts.
- To reduce the quantity of metal in hollow castings and hence reduce weight and material consumption.
- To obtain internal profiles with acceptable dimensional accuracy and surface finish.

Essential requirements of a good core

Requirement	Why it is necessary
Adequate green/dry strength	The core must withstand handling, baking, assembly and the force of molten metal.
High refractoriness	The core surface must resist the temperature of molten metal without fusing or softening.

Good permeability	Gases generated inside the core must escape; otherwise blowholes and gas defects may occur.
Collapsibility	After solidification the core should yield or break down so that casting contraction is not restrained.
Dimensional stability	The core must retain its correct size and shape during handling and pouring.
Smooth surface	Helps produce a clean internal surface of the casting.
Low gas evolution	Excess gas generation can cause porosity and blowholes.
Easy removal	The spent core should be removable without damaging the casting.

2. Types of Cores

Cores may be classified in several ways, but a common foundry classification is based on their position and manner of support in the mould.

Type of core	**Description and typical use**
Horizontal core	Placed horizontally, generally at the parting line. It is supported by core prints at its ends and is widely used for through holes with horizontal axes.
Vertical core	Placed vertically in the mould. Prints are provided in the cope and drag to locate and support the core.
Balanced core	Similar to a horizontal core but supported mainly from one end. A sufficiently long and strong core print is required to balance the unsupported portion. Commonly used for blind holes.
Drop core / wing core / stop-off core	Used when the cavity to be formed is not conveniently located on the mould parting line. It helps form projections, recesses or cavities

	above or below the normal parting surface.
Cover core	Used where the pattern may be moulded mainly in the drag and the core covers a portion of the mould from above. It often acts like a cover over the mould cavity.
Hanging core	Suspended from the cope and not supported from below. Because molten metal produces an upward buoyancy force, secure support is especially important.

Important: The names horizontal, vertical, balanced, drop, cover and hanging indicate how the core is placed or supported in the mould. The core print design must match the type of core.

3. Core Sand

Definition. Core sand is a specially prepared sand mixture used for making cores. Compared with ordinary moulding sand, core sand normally requires higher strength after drying, good permeability, high refractoriness and good collapsibility.

Typical constituents of core sand

Constituent	Function
Silica sand	Main refractory material and body of the core.
Binder	Bonds the sand grains and gives the core sufficient green and/or dry strength.
Water or activating liquid	Used where required to activate or distribute the binder and provide workability.
Core additives	Improve collapsibility, surface finish, strength, gas behaviour or resistance to metal penetration, depending on the application.

Common binders used for core making

- Organic binders: linseed oil and other drying oils, dextrin, molasses, cereal/flour-type binders and synthetic resins.
- Inorganic binders: sodium silicate, clay and cement-type binders for suitable applications.

- The choice of binder depends on the metal being poured, core size, required strength, production rate, method of hardening and desired collapsibility.

Properties required in core sand

Strength: The rammed core must withstand stripping from the core box, handling, baking, mould assembly and metal pressure.

Permeability: Gases must pass through the core and escape through vents or porous sand.

Refractoriness: The core should not fuse, soften or react excessively with molten metal.

Collapsibility: The core should collapse after metal solidifies so that the casting can contract freely.

Flowability: Sand should pack properly into corners and intricate portions of the core box.

Low gas evolution: The binder system should not produce excessive gases on heating.

Surface stability: The surface should resist washing and erosion by flowing molten metal.

General procedure for making a sand core

1. Prepare and mix the core sand with the selected binder/additives.
2. Apply a suitable parting material to the core box if required.
3. Fill the core box and ram the sand uniformly.
4. Provide reinforcement, core wires or rods for weak/large cores where necessary.
5. Provide vents so that gases can escape.
6. Strike off excess sand and carefully remove the core from the box.
7. Dry or bake the core, or chemically harden it, according to the binder system.
8. Dress and repair the core; join separate halves when required.
9. Apply core wash/coating where a smoother surface or extra refractoriness is needed.
10. Inspect the core for cracks, dimensional errors and adequate strength before placing it in the mould.

4. Core Prints

Definition. A core print is an additional projection provided on the pattern. During moulding it produces a recess or seat in the mould into which the core is located and supported.

Functions of core prints

- Locate the core accurately in the mould.
- Support the weight of the core before pouring.
- Resist the upward buoyancy force acting on the core after molten metal enters the mould.
- Prevent displacement of the core due to metal flow.
- Maintain the correct wall thickness and internal dimensions of the casting.

Buoyancy concept. When molten metal surrounds a core, the core experiences an upward force equal approximately to the weight of the molten metal displaced, minus the weight of the core. Therefore, core prints must have enough bearing area and strength. For difficult cases, chaplets may also be used as auxiliary supports.

Design considerations for core prints

- Core weight and size.

- Direction and magnitude of buoyancy force.
- Length and diameter/area of the print.
- Amount of taper needed for easy mould withdrawal and assembly.
- Accuracy required in locating the core.
- Strength of moulding sand around the print seat.
- Need for venting through or around the print.

5. Core Box

Definition. A core box is a pattern-like device containing the cavity into which core sand is packed to produce a core of the required shape and dimensions.

Core boxes may be made from wood, metal, plastic or other suitable materials. Wooden boxes are economical for small production, while metal boxes are preferred for greater durability, dimensional accuracy and repeated production.

Common types of core boxes

Core box type	Construction / use
Half core box	Produces one half of a symmetrical core. Two identical halves are made separately and then joined to form the complete core.
Split core box	Made in two matching parts, usually aligned by dowels. The complete core is formed in the assembled box and removed by separating the box halves.
Dump or slab core box	A simple box generally used for slab-like, rectangular or uncomplicated cores. The rammed core can be dumped or removed as a complete piece.
Strickle core box / strickle method	Used for large or irregular cores where a full core box would be expensive. A shaped strickle board sweeps the sand to the required contour.
Loose-piece core box	Contains removable pieces to permit withdrawal of a core having projections, recesses or other shapes that would otherwise lock in a rigid box.

Requirements of a good core box

- Correct dimensions and allowance for the required core.
- Smooth internal surfaces for good core finish.
- Adequate strength and rigidity.

- Easy filling, ramming and venting.
- Easy removal of the core without damage.
- Positive alignment for split boxes.
- Resistance to wear for the expected production quantity.

6. Melting Equipment - Types

Foundry melting equipment is selected according to the metal or alloy, quantity to be melted, required temperature and composition control, energy source, production rate and cost.

Melting equipment	Main features / typical use
Cupola furnace	Vertical shaft furnace fired mainly with coke; traditionally used for continuous melting of cast iron. High melting rate and economical for large quantities.
Crucible furnace	Metal is melted in a refractory crucible heated externally. Suitable for non-ferrous alloys and small/medium batches.
Reverberatory furnace	Flame and hot gases heat the metal by radiation/reflection from the roof; the fuel does not pass directly through the metal charge in the same way as in a cupola.
Rotary furnace	Rotating refractory-lined shell gives good mixing and heat transfer. Used for several ferrous/non-ferrous melting applications.
Electric arc furnace	Heat is generated by electric arcs. Capable of very high temperatures and useful where composition control and steel/iron melting are required.
Induction furnace	Electromagnetic induction heats and stirs the charge. Clean, fast and provides good composition/temperature control; widely used in modern foundries.
Resistance furnace	Electrical resistance heating; mainly used for smaller non-ferrous melting/holding operations where clean heating is desired.

Exam point: For traditional cast-iron foundry practice, the cupola is one of the most important

furnaces because it can melt iron continuously and at a high rate.

7. Cupola Furnace - Construction

Definition. A cupola is a vertical, cylindrical, refractory-lined shaft furnace used primarily for melting cast iron. Solid charge is introduced from the top while air is blown through tuyeres near the lower part of the shaft. Coke supplies heat and also participates in the furnace atmosphere.

Main parts of a cupola

Part	Function
Steel shell	Forms the outer cylindrical body of the furnace.
Refractory lining	Protects the shell from high temperature and provides thermal insulation.
Bottom doors / drop bottom	Hinged doors support the temporary sand bottom during operation and can be dropped after the heat for cleaning.
Sand bottom / hearth	Sloped refractory/sand bed that directs molten iron toward the tap hole.
Well or crucible region	Space below the tuyeres where molten metal collects before tapping.
Tuyeres	Openings through which the air blast enters the furnace.
Wind belt / wind box	Distributes air from the blower uniformly around the tuyeres.
Blower	Supplies the forced air needed for coke combustion.
Tap hole	Opening through which molten iron is removed.
Slag hole	Opening located above the tap hole for removing slag.
Charging door	Opening in the upper shaft through which coke, metal and flux are charged.
Stack / chimney	Upper extension that carries furnace gases away

	and provides the shaft for preheating descending charge.

8. Cupola Furnace - Working

A. Preparation before melting

11. Clean the furnace after the previous heat and inspect/repair the refractory lining.
12. Close and support the bottom doors.
13. Prepare a sloping sand bottom so that molten metal can flow toward the tap hole.
14. Close the tap hole and slag hole initially with suitable refractory clay/sand plugs.
15. Charge a coke bed of sufficient height above the tuyeres.
16. Ignite the coke bed and allow it to burn until it is fully incandescent.

B. Charging and melting

17. Start the air blast through the tuyeres.
18. Charge alternate layers of metal, coke and flux through the charging door.
19. Coke burns intensely in front of and above the tuyeres and develops the heat required for melting.
20. Hot gases rise through the shaft and preheat the descending charge.
21. As the metal reaches the melting zone it melts and droplets descend through the hot coke bed.
22. Limestone or another suitable flux combines with ash and impurities to form slag.
23. Molten iron collects in the well below the tuyeres, while the lighter slag floats above it.

C. Tapping and shutdown

24. Open the slag hole periodically to remove accumulated slag.
25. Open the tap hole and collect molten iron in a ladle for pouring into moulds.
26. Maintain charging and blast if continuous melting is required.
27. At the end of the heat, stop charging and finally stop the blast after the remaining charge is melted.
28. Drain metal and slag, then open/drop the bottom doors so coke and residual material can fall out for cleaning.

Key idea: A cupola operates on the counter-current principle: solid charge moves downward while hot gases move upward. The rising gases preheat the charge, improving thermal utilization.

9. Zones of a Cupola Furnace

The cupola shaft is commonly divided into characteristic zones according to the main reactions and temperature conditions occurring at different heights.

Zone	Approximate location	Main events

Well / crucible zone	Below the tuyeres	Molten iron and slag collect here before tapping. This is mainly a collection region rather than a combustion region.
Combustion / oxidizing zone	At and immediately above the tuyeres	Oxygen from the blast reacts strongly with coke. The principal heat-producing reaction is $C + O_2 \rightarrow CO_2$. This region develops the highest furnace temperatures.
Reducing zone	Above the combustion zone	CO_2 passing through hot coke reacts further: $CO_2 + C \rightarrow 2CO$. The resulting carbon monoxide creates a reducing atmosphere and limits oxidation of the charge.
Melting zone	Above the reducing zone	Metal reaches its melting temperature, melts and trickles downward as droplets. It absorbs substantial sensible and latent heat.
Preheating / charging zone	Upper part of shaft below/around charge level	Rising hot gases heat the incoming metal, coke and flux before they descend to the hotter zones. Moisture is driven off and the charge temperature rises.
Stack / chimney region	Above the charging level	Waste gases leave the furnace; the stack provides draft path and safe discharge of gases.

Important chemical reactions

- Combustion: $C + O_2 \rightarrow CO_2$ + heat. This is strongly exothermic and supplies much of the melting heat.
- Reduction/Boudouard reaction: $CO_2 + C \rightarrow 2CO$. This consumes heat but produces a reducing gas.
- Flux decomposition (for limestone): $CaCO_3 \rightarrow CaO + CO_2$. The CaO combines with acidic ash/impurities to help form slag.

Note on exact temperatures: The precise temperature boundaries of cupola zones vary with furnace size, coke ratio, air blast, charge and operating practice. For examination purposes, understanding the sequence and dominant reactions is more important than memorizing a single temperature value for every zone.

10. Cupola Efficiency and Performance

Cupola performance is judged by how effectively fuel is used to melt and superheat the iron at the required rate and quality.

10.1 Melting ratio

A simple and widely used fuel-utilization index is the melting ratio:

$$\textbf{Melting ratio} = \textbf{Weight of metal melted} / \textbf{Weight of coke consumed}$$

A higher melting ratio generally means more metal is melted per unit mass of coke, provided the required metal temperature and quality are maintained.

10.2 Thermal efficiency

Thermal efficiency may be expressed as the useful heat absorbed by the metal divided by the heat supplied by the fuel:

$$\eta = (\text{Useful heat gained by metal} / \text{Heat supplied by coke}) \times 100\%$$

The useful heat gained by the metal includes the heat required to raise the solid charge to its melting point, the latent heat of fusion, and the heat required to superheat the liquid metal to the tapping temperature.

10.3 Factors affecting cupola efficiency

- Correct coke-to-metal ratio: too little coke gives low temperature; excessive coke wastes fuel and may alter chemistry.
- Correct bed-coke height above the tuyeres.
- Adequate and uniformly distributed air blast.
- Proper size and distribution of coke and metal pieces to maintain permeability of the charge bed.
- Good refractory condition and reduced heat loss through the shell.
- Proper charging practice and steady operation.
- Effective preheating of incoming charge by outgoing gases.
- Suitable flux quantity: excessive flux increases slag volume and heat consumption.
- Control of air leakage and correct tuyere condition.
- Clean, reasonably dry charge to avoid unnecessary heat use and gas generation.

10.4 Major heat losses

- Heat carried away by hot exhaust gases.
- Heat absorbed by slag.
- Heat lost through the furnace wall and openings.
- Heat used in endothermic reactions and moisture evaporation.
- Heat remaining in unburnt or partially utilized fuel and discharged material.

Cupola Furnace - Advantages and Limitations

Advantages	Limitations
High melting rate and continuous operation.	Composition control is less precise than in many electric furnaces.
Simple construction and comparatively low operating cost for large cast-iron production.	Coke combustion introduces gases and can change carbon/sulphur content.
Can use a variety of cast-iron/steel scrap and returns.	Air pollution and dust/fume control are important environmental concerns.
Good thermal contact between hot gases and descending charge.	Frequent lining maintenance and careful coke/charge control are required.
Molten metal can be tapped repeatedly during a heat.	Not equally suitable for every alloy or for very small clean melts.

11. Quick Revision

Term	One-line recall
Core	Sand body placed in a mould to form internal cavities or difficult surfaces.
Core print	Pattern projection that creates a seat to locate and support a core.
Core box	Tool used to shape and produce a sand core.
Core sand	Special sand mixture designed for core strength, permeability, refractoriness and collapsibility.
Cupola	Vertical coke-fired shaft furnace mainly used for melting cast iron.
Tuyere	Opening through which air blast enters the cupola.
Well	Region below tuyeres where molten iron collects.
Combustion zone	Zone where coke burns with oxygen and major heat is generated.

Reducing zone	Zone where CO_2 reacts with hot coke to form CO.
Melting zone	Zone in which metal charge melts.
Preheating zone	Upper zone where rising hot gases heat the descending charge.
Melting ratio	Mass of metal melted per unit mass of coke consumed.

Short Viva / Examination Questions

29. Define a core. Why are cores used in casting?
30. List the important properties required of core sand.
31. Differentiate between a horizontal core and a balanced core.
32. What is a core print? Why is it necessary?
33. What is meant by buoyancy force on a core?
34. What is a core box? Name four types of core boxes.
35. Why is a split core box provided with alignment features such as dowels?
36. Classify common foundry melting furnaces.
37. Draw a neat sectional sketch of a cupola and label its main parts.
38. Explain the construction and working of a cupola furnace.
39. Name and explain the zones of a cupola furnace.
40. Write the important reactions in the combustion and reducing zones.
41. What is the purpose of limestone/flux in cupola melting?
42. Define melting ratio and thermal efficiency of a cupola.
43. List factors that improve cupola efficiency.

Reference

Primary reference framework: Hajra Choudhury, S. K.; Hajra Choudhury, A. K.; and Nirjhar Roy, S. K., Elements of Workshop Technology, Volume I - Manufacturing Processes, Media Promoters & Publishers.

The wording in these notes is original and condensed for learning. Terminology and topic organization follow standard workshop/foundry technology treatment consistent with the cited reference.